Progress in Nuclear Energy 160 (2023) 104679

Contents lists available at ScienceDirect

Progress in Nuclear Energy

journal homepage: www.elsevier.com/locate/pnucene

Development and verification of helium–xenon mixture cooled small reaction system

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ARTICLE INFO

Keywords:
Helium-xenon mixture
Space reactor
Brayton cycle
Systems analysis

ABSTRACT

Although Helium–Xenon mixture cooled small reactor has been widely concerned, there is no mature thermal calculation software that can be applied to Helium–Xenon mixture cooled small reactor in China. In this paper, a semi empirical formula used to predict the physical properties of binary inert gas mixtures is used to establish a physical property calculation model of Helium–Xenon mixture gas mixtures, and the flow and heat transfer calculation model, turbine model and compressor model are modified and improved, so as to achieve the calculation of the Brayton cycle system. The function of the original system analysis program is expanded and improved, and the simulation program for he xenon mixed gas reactor system is developed. At the same time, the closed Brayton cycle system is established by using the system analysis program developed in this paper, and the calculation results are compared with the design values. The maximum relative error is 6.9%. The steady-state operation results of the model are in good agreement with the design operation parameters, which proves the applicability of the improved simulation model. The steady-state response characteristics analysis, the introduction of positive and negative reactivity accidents and the calculation of transient conditions such as step load drop of the reactor were carried out. The overall trend is the same as the literature survey results. The results show that the system analysis program developed in this paper is suitable for the analysis of the Brayton cycle system of Helium–Xenon mixture cooled small reactor.

1. Introduction

Human exploration of space has never stopped, and the space field is an important strategic area for the development of many countries. In the future, if space technology wants to meet the requirements of high energy density, high power level and long mission cycle, nuclear reactor power supply is the inevitable trend of space technology development (Iaea, 2005; Zhang et al., 2020; Qin et al., 2021a). EL-GENK et al. (EL-GENK and TOURNIER, 2007a) compared the physical parameters of helium neon, helium argon, helium krypton and Helium–Xenon (He–Xe) mixtures in detail, and found that adding only a small amount of xenon can effectively increase the molar mass of the working medium. At the same time, the He–Xe mixture has a greater heat transfer coefficient, which has a positive impact on the thermodynamic performance of the system, and can effectively reduce the number of compressor and turbine stages, so that it can meet the requirements of aeroengine for the number of turbomachinery stages not exceeding seven. Therefore, the He–Xe mixture is suitable for use as the coolant of the thermal cycle of space reactors.

Since 2006, the United States has successively carried out "the Prometheus Program" (JIMO) (MJ and MJ, 2006; Ashcroft and Eshelman, 2007; Taylor, 2005), the conceptual design of Submerged - Subcritical safe space reactor (S^4), etc. (Schriener and El-Genk, 2011; Schillo et al., 2016). The above-mentioned reactors use He–Xe mixture as coolant. EL-GENK et al. (EL-GENK and PARAMONOV, 1994) developed transient characteristic analysis and safety characteristic analysis for SP-100 system and TOPAZ-II system by using FORTRAN language. And based on the space reactor power system of submerged subcritical reactor (S^4),

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https://doi.org/10.1016/j.pnucene.2023.104679
Received 27 November 2022; Received in revised form 15 February 2023; Accepted 30 March 2023
Available online 15 April 2023
0149-1970/© 2023 Published by Elsevier Ltd.

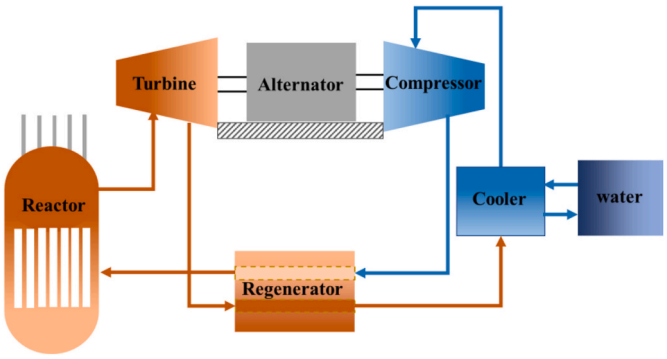


Fig. 1. Schematic diagram of Brayton cycle system
(Note: The main equipment in the Fig. 1 are follows: (1)Reactor (2)Turbine (3) Regenerator (4)Cooler (5)Compressor (6)Alternator).

they established a dynamic simulation model of the space reactor power system with multiple closed Brayton cycle (CBC) energy conversion loops, and demonstrated its startup transient. Wright et al. (Wright, 2003; Wright and Sanchez, 2005) used the SIMULINK platform to conduct dynamic simulation analysis on the closed Brayton cycle system using He–Xe mixture as coolant, and introduced control on the basis of existing programs to study the response characteristics of the system under different control methods. A large number of studies on gas cooled micro reactor system analysis program have been carried out in China, but most of them are aimed at reactors whose cooling working medium is supercritical carbon dioxide. Ming Yang et al. (Ming et al., 2022) used Simulink program to establish system analysis program for supercritical carbon dioxide combined with small Brayton cycle reactor system. They fully consider the physical property changes of working fluid under variable working conditions and establish an accurate physical property database of working fluid. Successfully carry out the steady-state simulation and transient dynamic simulation of small Brayton cycle system. Zhang et al. (Zhang et al. et al.) modified RELAP5 and conducted benchmark tests through typical designs. Without the intervention of control or auxiliary system, transient conditions, including reactor power step reduction, power load loss and cooling water temperature disturbance, were simulated to study the natural response of the SCO_2-Cooled Reactor System.

Although helium-xenon gas-cooled small reactors are widely concerned at present, Qin et al. (2020) and Zhang et al. (2021) have successively carried out the conceptual design and safety analysis of open gas-cooled space reactors, and the development of closed space gas-cooled reactor system analysis program, which provides a useful reference for the design of megawatt space nuclear systems and the analysis of thermal-hydraulic safety characteristics. However, there is no mature system analysis program that can be applied to He–Xe mixture cooled small reactors. Therefore, the development of helium-xenon gas-cooled small reactor system analysis procedures is needed to compensate for the shortcomings of existing research. Therefore, it is necessary to develop the system analysis program of He–Xe mixture cooled small reactor to make up for the lack of domestic research. RELAP5 is a calculation program approved by the U.S. Nuclear Regulatory Commission and applied to the safety assessment of large nuclear reactors. This procedure is developed for water reactors and is not applicable to reactors with He–Xe mixture as coolant (DAVIS, 2006). At present, the application of this program is relatively mature, with fast calculation speed and high accuracy, which has been widely recognized by the nuclear industry. At the same time, there is little difference between the calculation steps and methods of pressurized water reactor and He–Xe mixture cooled small reactor. The main differences are the physical parameters of working fluid, the models of compressor and gas turbine. Idaho National Laboratory (Guillen et al., 2006; Fisher et al., 2005; Ravindran et al., 2010) of the United States upgraded RELAP5-3D version through the structural reorganization of RELAP5-3D computing mode, and adding the physical model required by the fourth generation reactor modeling to the original model, it is transformed into a system analysis program suitable for the fourth generation small reactor. McCann et al. (McCann, 2007) used RELAP5-3D to establish the space reactor power system proposed by Prometheus project, and carried out the steady-state and transient operation characteristics analysis of the reactor with He–Xe mixture combined with closed Brayton cycle.

Through the above analysis, it can be seen that it is a convenient method to develop the analysis program of He–Xe mixture cooled small reactor system to adapt RELAP5 program to He–Xe mixture cooled small reactor. Therefore, based on the original system analysis program, this paper establishes a high-precision physical property calculation model of He–Xe mixture, so as to reduce the error brought by calculating the physical property of He–Xe mixture. At the same time, the flow and heat transfer model and the turbine model in the program are modified, and the compressor model is introduced to realize the calculation of the Brayton cycle system.

2. Model overview and implementation

Closed Brayton cycle system is adopted for He–Xe mixture cooled small reactor. The system is mainly composed of reactor, gas turbine, regenerator, cooler and compressor. The schematic diagram of Brayton cycle system is shown in Fig. 1. The He–Xe mixture is heated in the reactor. The high-temperature and high-pressure He–Xe mixture enters the gas turbine and expands in the gas turbine. The working medium flowing out from the gas turbine enters the hot side of the regenerator, is cooled by the cooler, and is compressed in the compressor. Then it flows into the regenerator for cold measurement, and flows into the core after preheating at the hot side of the regenerator, forming a complete closed Brayton cycle. The hot side of the regenerator recovers the residual heat from the exhaust gas of the gas turbine to preheat the working medium flowing out of the compressor outlet, improve the physical parameters of the working medium at the reactor inlet, and thus improve the thermal cycle efficiency. The compressor and the generator are driven by the gas turbine at the same time. The compressor compresses the low-temperature and low-pressure He–Xe mixture at the outlet of the cooler and provides the driving head of the circulation circuit. The physical property calculation model, flow and heat transfer calculation model, main equipment model and its implementation in the system analysis program of He–Xe mixture will be described in detail below.

The hot side of the regenerator recovers the residual heat from the exhaust gas of the gas turbine to preheat the working medium flowing out of the compressor outlet, improve the physical parameters of the working medium at the reactor inlet, and thus improve the thermal cycle efficiency.

2.1. Physical property calculation model of He–Xe mixture

2.1.1. Mathematical model

The physical property calculation model of He–Xe mixture in original system analysis program is based on Sutherlands law and ideal gas equation of state. Sutherlands model was proposed by Australian physicist William Sutherland in 1893 (Sutherland, 1893). Based on the theory of gas dynamics, this model expounds the relationship between the dynamic viscosity and absolute temperature of ideal gas. The research of El Genk et al. (EL-GENK and TOURNIER, 2007b) shows that using Sutherlands' law to solve the physical properties of He–Xe mixture will introduce large errors, which is not suitable for the calculation of He–Xe mixture. The dilution gas theory based on Chapman – Enskog method is also not applicable to the working range of space reactor coolant (Mason and Uribe, 1996). Therefore, a more accurate calculation method is needed to calculate the physical properties of He–Xe mixture. Tournier et al. (2006) proposed a semi empirical formula to predict the physical properties of binary inert gas mixtures based on the

Table 1
Virial coefficient expression.

Virial coefficient	Calculation formula
Second (except helium)	$B = \psi_B(\theta) = v^*\left[-102.6 + \left(102.732 - 0.001 \times \theta - \frac{0.44}{\theta^{1.22}}\right) \times \tanh(4.5\sqrt{\theta})\right]$
Second (helium)	$B_{He}[\mathrm{cm}^3/\mathrm{mol}] = 8.4 - 0.0018T + \frac{115}{\sqrt{T}} - \frac{835}{T}$
Second (mixed gas)	$B = x_1^2 B_{11} + 2x_1 x_2 B_{12} + x_2^2 B_{22}$ $B_{12} = v_{12}^* \cdot \psi_B(T/T_{cr,12})$ $B_{11} = v_{11}^* \cdot \psi_B(T/T_{cr,11})$ $B_{22} = v_{22}^* \cdot \psi_B(T/T_{cr,22})$
Third	$C = v^{*2}\left[0.0757 + \left(-0.0826 - 3.6 \times 10^{-5}\theta + \frac{0.0237}{\theta^{0.059}}\right) \times \tanh(0.84\theta)\right]$
Third (mixed gas)	$C = x_1^3 C_{111} + 3x_1^2 x_2 C_{112} + 3x_1 x_2^2 C_{122} + x_2^3 C_{222}$ $C_{112} = (C_1^2 C_2)^{\frac{1}{3}}$ $C_{122} = (C_1 C_2^2)^{\frac{1}{3}}$

Note: $\theta = T/T_{cr}$ is relative temperature; T_{cr} is critical temperature of gas; v^* is the critical specific volume of gas, v_{12}^* is the average specific volume of mixed working medium, and subscripts 1 and 2 represent helium and xenon respectively; $T_{cr,12}$ is the average critical temperature of the mixed working medium.

principle of corresponding states on the basis of a large number of previous experimental data, and studied the physical properties of different inert gas mixtures with a pressure range of 0.1 MPa–20 MPa and a temperature lower than 1400 K. Based on the semi empirical formula proposed by Tournier et al. Biao Zhou et al. (ZHOU et al., 2020) developed the physical property calculation program of He–Xe mixture in FORTRAN language, and verified the accuracy of the calculation method.

In this paper, the physical properties of He–Xe mixture are calculated according to the semi empirical formula proposed by Tournier et al. (2006). Some calculation formulas are as follows:

The density of He–Xe mixture is calculated by formula (1):

$$\rho = \frac{p}{ZR_gT} \tag{1}$$

Where, ρ is the density of He–Xe mixture, kg/m^3; R_g is the gas constant of He–Xe mixture working medium; Z is the actual gas compression factor; T is the temperature of He–Xe mixture, K; P is the pressure of He–Xe mixture, Pa.

The compressibility factor Z is assumed a second order polynomial of the molar density, thus:

$$Z = 1 + \frac{B}{v} + \frac{C}{v^2} \tag{2}$$

Where, v is the specific volume of gas, m^3/kg. B, C drepresent the second virial coefficients, third virial coefficients. The second virial coefficient represents the interaction between two molecules, and the third virial coefficient represents the interaction between three molecules. Using the combination of mathematical analysis and experimental data, the empirical formula of virial coefficient of actual gas can be derived (March 1970). The empirical formula of virial coefficient of ideal gas and actual gas is quite different, so it needs to be calculated separately. In this paper, the second and third order virial coefficients are used to analyze helium xenon mixtures with reference to Tournier et al. (2006) formula for calculating the intermolecular potential correlation between pure inert gases and inert gas mixtures. The calculation formula of partial virial coefficient is shown in Table 1.

When calculating the dynamic viscosity and thermal conductivity of the mixed working medium, the corresponding state principle and experimental data are used (Tournier et al., 2006). The viscosity expression of He–Xe mixture is expressed as:

$$\mu_{mix} = \mu^0 + \left(1 - \frac{1}{2.3}\right)\mu^* \times \psi_\mu\left(\frac{0.291 \times v^*}{M}\rho\right) \tag{3}$$

Where, μ^0 is the weighted average value of viscosity, and the latter half is the actual gas correction. $\psi_\mu(\rho_r)$ is the fitting function of test data, μ^* is the fitting viscosity, and v^* is the average critical specific volume.

The thermal conductivity of He–Xe mixture is expressed as:

$$\lambda_{mix} = \lambda^o + \left(1 - \frac{1}{2.94}\right)\lambda^* \times \Psi_\lambda\left(\frac{0.291 \times V^*}{M}\rho\right) \tag{4}$$

Where, λ^o is the weighted average value of thermal conductivity, and the second half is the actual gas correction. $\Psi_\lambda(\rho_r)$ is the fitting function of test data, and λ^* is the fitting thermal conductivity.

It is found by calculation that when solving the specific heat capacity of He–Xe mixture, the virial coefficient of He–Xe mixture needs to be derived many times, and the process is extremely cumbersome. In this paper, the constant pressure specific heat capacity data of He–Xe mixture at different temperatures and pressures under different mixing ratios are taken to draw curves, as shown in Fig. 2. It can be seen from the figure that the constant pressure specific heat capacity of He–Xe mixture has little correlation with temperature and pressure. With the change of temperature and pressure, the specific heat capacity at constant pressure of He–Xe mixture is almost unchanged under the same He–Xe mixture ratio. The constant pressure specific heat capacity of

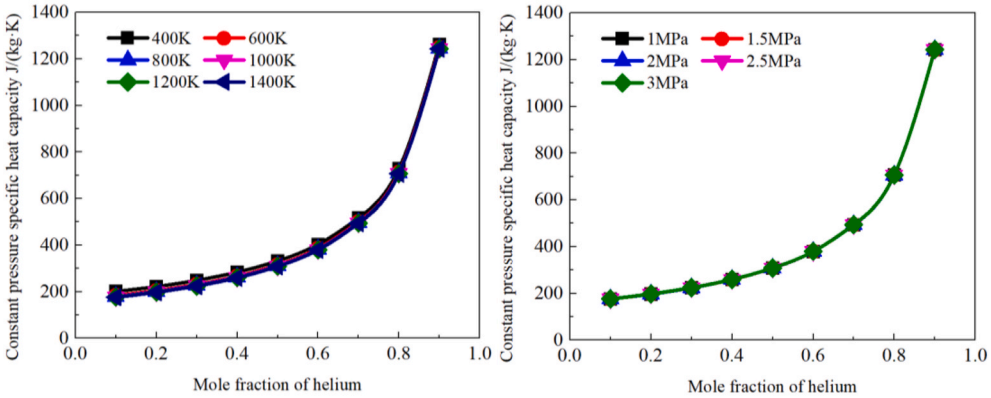


Fig. 2. Schematic diagram of variation of specific heat capacity at constant pressure with molar mass fraction of helium.

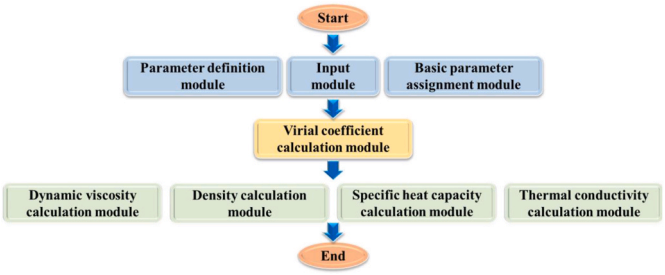


Fig. 3. Flow chart of physical property calculation of He–Xe mixture.

He–Xe mixture only changes with the molar mass fraction of helium. Therefore, this paper uses the polynomial fitting method to solve the constant pressure specific heat of He–Xe mixture under different molar fractions of helium. The fitting results are shown in formula (5).

$$C_p = 2.6276 \times x_1^5 - 5.3238 \times x_1^4 + 4.1073 \times x_1^3 - 1.4134 \times x_1^2 + 0.2345 \times x_1 + 0.0071 \quad (5)$$

Where, x_1 is the mass fraction of helium in He–Xe mixture.

2.1.2. Program development and verification

In this paper, referring to the semi empirical formula proposed by Tournier et al. (2006), FORTRAN 77 language is used to develop the physical property calculation program of He–Xe mixture. When the temperature is 500–1700 K and the pressure is 1–5 MPa, the insulation coefficient is taken as 1.67. The specific calculation process is shown in Fig. 3. In this calculation module, the density, specific heat capacity, viscosity, thermal conductivity and Prandtl number of He–Xe mixture can be obtained by inputting temperature, pressure and helium mole fraction.

When the mass fraction of helium is 0.07, that is, the relative molecular mass of He–Xe mixture is 40 g/mol, the calculated results are compared with the physical property parameters of the existing He–Xe mixture working medium, it is necessary to consider the influence of temperature and pressure changes on the physical properties when analyzing the calculation accuracy of the physical property calculation program. First, fix the temperature at 500 K, and take 7 pressure nodes: 1, 1.5, 2, 2.5, 3, 3.5, 4 MPa; Then fix the pressure to 1 MPa and take 7 temperature nodes: 500, 550, 600, 650, 700, 750, 800 K.

The comparison results are shown in Fig. 3 and Fig. 4. It can be seen from the figure that the calculated results of the program are very consistent with the physical parameters of the He–Xe mixture calculated in reference (Meng et al., 2020). The maximum error of He–Xe mixture density calculation is 5.7%, the maximum error of dynamic viscosity calculation is 2.25%, and the maximum error of tvery consistent with the physical parameters of the He–Xe mixture calculated in referencehermal conductivity calculation is 3.94%. The error of the whole calculation program is less than 5.7%, indicating the accuracy of the physical property calculation program.

2.2. Flow heat transfer model

In original system analysis program, Dittus-Boelter (Dittus and Boelter, 1985) formula is mostly used for forced convection heat transfer calculation. Hao et al. (Qin et al., 2021b)carried out a numerical study on the heat transfer characteristics of helium and xenon gas mixture in uniformly heated circular and 3-Pl channels. Taylor et al. (1988) studied the internal forced convection heat transfer relationship of gas mixture under low Prandtl number, and explored the applicability of various heat transfer formulas for the flow heat transfer calculation of He–Xe mixture. Through experiments, it is found that when the Prandtl number is low, the extrapolation of Dittus-Boelter formula will seriously overestimate the heat transfer parameters, resulting in the calculated wall temperature being far lower than the actual value, which brings about

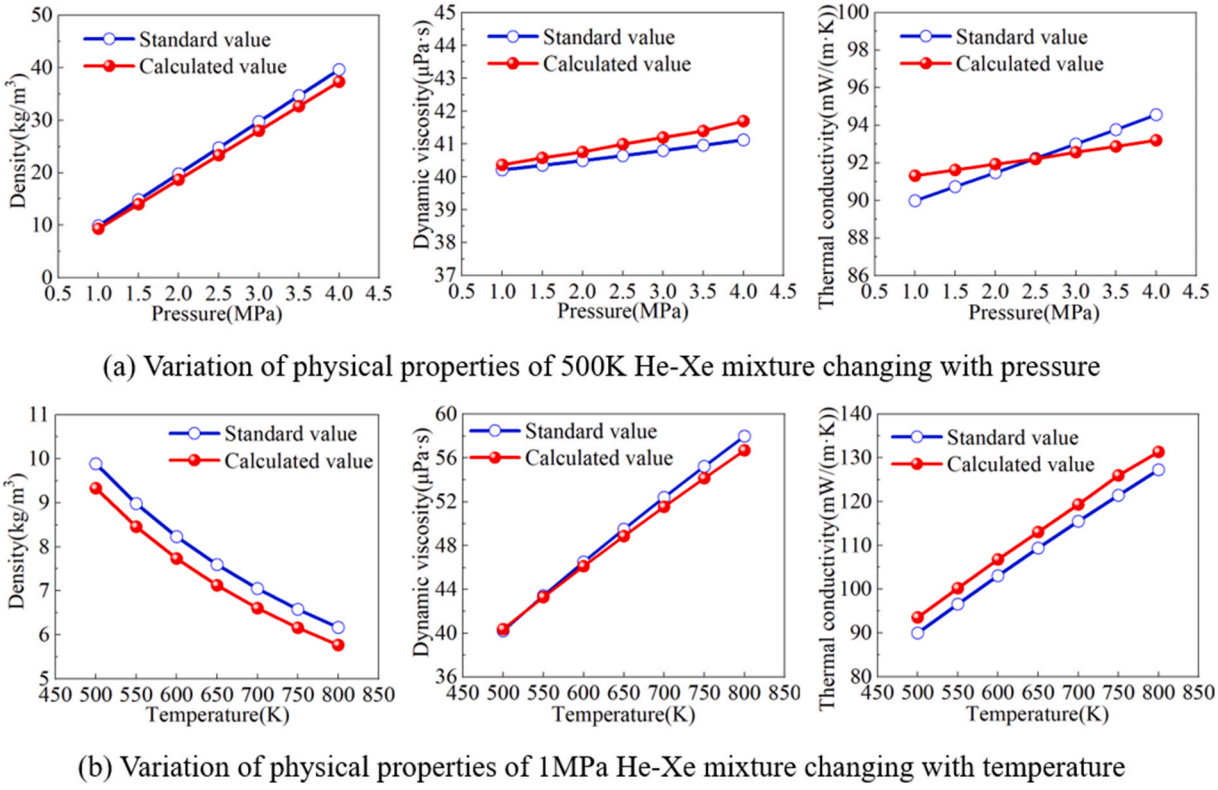


Fig. 4. a Variation of physical properties of 500 K He–Xe mixture changing with pressure.
b Variation of physical properties of 1 MPa He–Xe mixture changing with temperature.

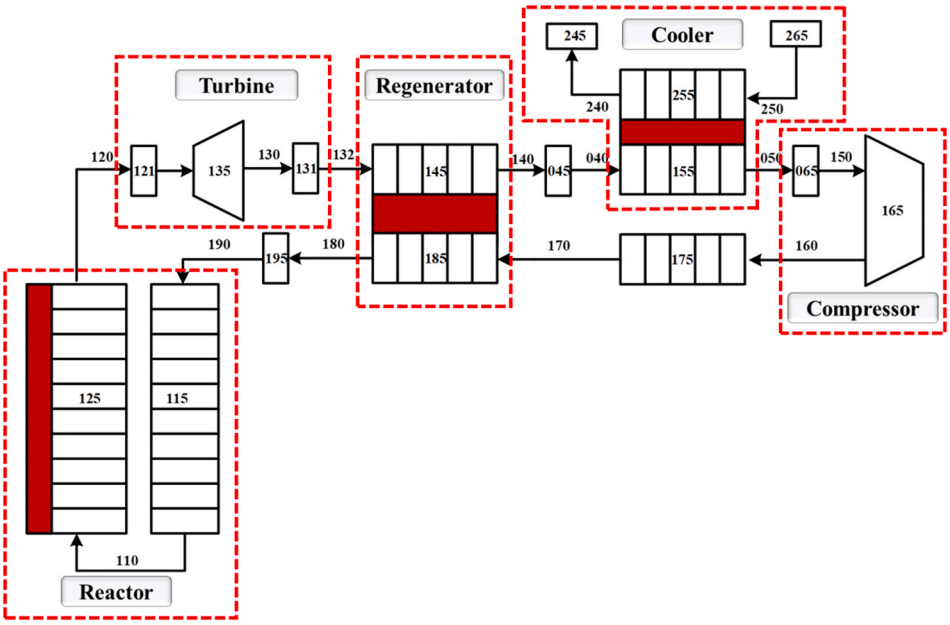


Fig. 5. Brayton cycle system node diagram.

Table 2
Verification of operating parameters of Brayton cycle system.

Variable name	Design value	Calculated value	Error
Core inlet temperature (K)	907.00	901.90	0.56%
Core inlet pressure (MPa)	1.41	1.35	4.26%
Core outlet temperature (K)	1200.00	1210.30	−0.86%
Core outlet pressure (MPa)	1.40	1.35	3.85%
Gas turbine outlet temperature (K)	966.30	959.50	0.70%
Gas turbine outlet pressure (MPa)	0.73	0.72	1.63%
Outlet temperature of hot section of regenerator (K)	553.70	541.19	2.26%
Outlet pressure of hot section of regenerator (MPa)	0.70	0.71	−1.86%
Outlet temperature of hot section of cooler (K)	340.00	340.32	−0.09%
Outlet pressure of cooler hot section (MPa)	0.67	0.70	−4.95%
Compressor outlet temperature (K)	501.60	496.79	0.96%
Compressor outlet pressure (MPa)	1.41	1.40	1.01%
Outlet temperature of cold section of regenerator (K)	907.00	896.19	1.19%
Outlet pressure of regenerator cold section (MPa)	1.41	1.37	2.49%
Mass flow (kg/s)	11.2	11.27	0.65%
Compressor pressure ratio	2.11	2.00	5.21%
Gas turbine expansion ratio	1.911	1.86	2.47%

safety hazards. Therefore, when original system analysis program is used to develop the analysis program for He–Xe mixture cooled small reactor system, Taylor formula is selected to modify the original heat transfer relationship:

$$\begin{aligned} \mathrm{Nu}_b &= 0.023\mathrm{Re}_b^{0.8}\,\mathrm{Pr}^{0.65}(T_\mathrm{w}/T_\mathrm{b})^{-n} \\ n &= [0.57 - 1.59/(x/D)] \end{aligned} \tag{6}$$

Where, Re_b is the Reynolds number of variable physical properties; Pr is the Prandtl number; T_w is the wall temperature, K; T_b is the average temperature of helium xenon mixture, K; n is the variable physical property correction factor; x is the axial distance between the uniform heating section and the starting point of the heating section, m; D is the overall length of the heating section, m. The applicable range of the formula is $1.8 \times 10^4 < Re < 6 \times 10^4$, $T_\mathrm{w}/T_\mathrm{b} < 2$, where $T_\mathrm{w}/T_\mathrm{b}$ is the ratio of surface temperature to volume temperature, and x/D represents the ratio of the inlet distance of the test section to the inner diameter of the test section.

2.3. Turbine model

Original system analysis program already has the calculation ability of turbine equipment, but the model has some defects, that is, it is found that the work transferred from high temperature and pressure working fluid to the shaft is not equal to the actual energy decrease of the working fluid (Wright and Sanchez, 2005). Therefore, this paper modifies the existing model on the basis of the original turbine equipment program. By adding an appropriate source term to the turbine energy equation, the energy consumption of working fluid can be matched with the increase of shaft power.

For stable single-phase fluid, the energy equation can be written as:

$$(\dot{m}U)_j^{j+1} + \left(P\frac{\dot{m}}{\rho}\right)_j^{j+1} = \dot{Q} \tag{7}$$

Where, $\dot{m}$ is the mass flow rate of mixed working medium, kg/s; U is the specific internal energy, J/kg; j is the indicates the nozzle number; $\dot{Q}$ is the source term due to turbine model correction, J.

Therefore, formula (7) can be rewritten in the form of specific enthalpy for the inlet and outlet pipe interface with the same pressure as:

$$h_2 - h_1 + \frac{P_1 - P_2}{\rho_1} = \frac{\dot{Q}}{\dot{m}} \tag{8}$$

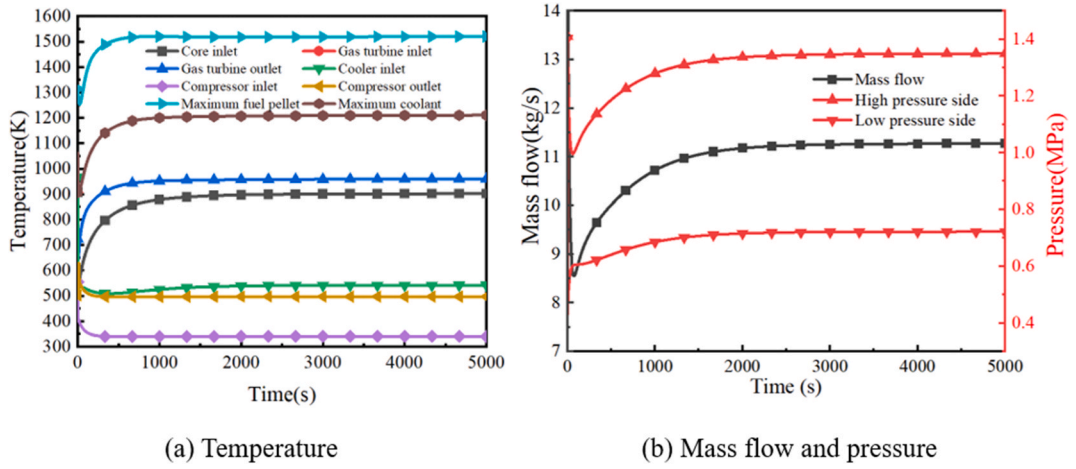
Where, h_1 is the actual enthalpy of gas turbine inlet, J/kg; h_2 is the actual enthalpy value at the outlet of the turbine, J/kg; P_1 is the inlet pressure, Pa; P_2 is the outlet pressure, Pa; ρ_1 is the inlet density, kg/m^3.

At the same time, the total energy equation of the turbine can be written as:

$$h_2 - h_1 + 0.5\left(v_2^2 - v_1^2\right) = -\frac{\dot{W}}{\dot{m}} = W \tag{9}$$

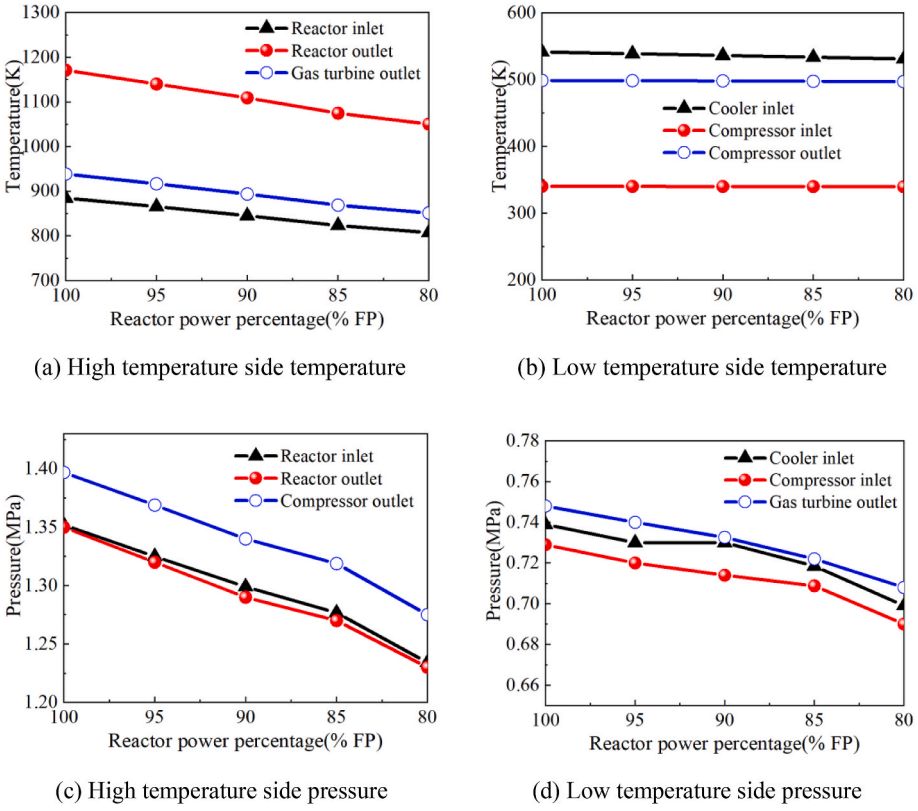
Where, v_1 is the inlet specific volume, m^3/kg; v_2 is the outlet specific volume, m3/kg; $\dot{W}$ is the total energy of the turbine, J; W is the total power of the turbine, J/kg.

Therefore, the source term can be obtained from the above two equations:



(a) Temperature (b) Mass flow and pressure

Fig. 6. Schematic Diagram of Key Parameters Changing with Time in Steady State Operation We have checked and corrected it in Line 3 in the revised manuscript on Page 16, as "It can be seen from Fig. 6 that it takes a long time for the system to achieve steady-state operation.



(a) High temperature side temperature (b) Low temperature side temperature

(c) High temperature side pressure (d) Low temperature side pressure

Fig. 7. Steady state response characteristics of closed Brayton cycle system with different reactor power.

$$\dot{Q} = \dot{m}\left[\frac{P_1 - P_2}{\rho_1} + 0.5\left(v_2^2 - v_1^2\right)\right] = -\dot{W} \tag{10}$$

The power of the shaft in the turbine equipment can be approximately written as:

$$\dot{W} = \frac{\eta(P_1 - P_2)}{0.5(\rho_1 + \rho_2)}\dot{m} \tag{11}$$

Where ρ_2 is the outlet density, kg/m^3. Among them, efficiency η is consistent with the calculation method of compressor adiabatic efficiency, that is, the ratio of actual power to isentropic power, that is:

$$\eta = \frac{\text{Actual power}}{\text{Isentropic power}} = \frac{(h_1 - h_2)\dot{m}}{(h_1 - h_{2'})\dot{m}} = \frac{h_1 - h_2}{h_1 - h_{2'}} \tag{12}$$

Where, $h_2{}'$ is the isentropic process enthalpy of working medium at the outlet of gas turbine, J/kg. The final simultaneous equations (10)–(12) can be obtained. The source terms that need to be added to the turbine energy equation are:

$$\dot{Q} = \dot{m}\left[\frac{P_1 - P_2}{\rho_1} + 0.5\left(v_2^2 - v_1^2\right)\right] - \frac{(h_1 - h_2)(P_1 - P_2)\dot{m}}{0.5(h_1 - h_{2'})(\rho_1 + \rho_2)} \tag{13}$$

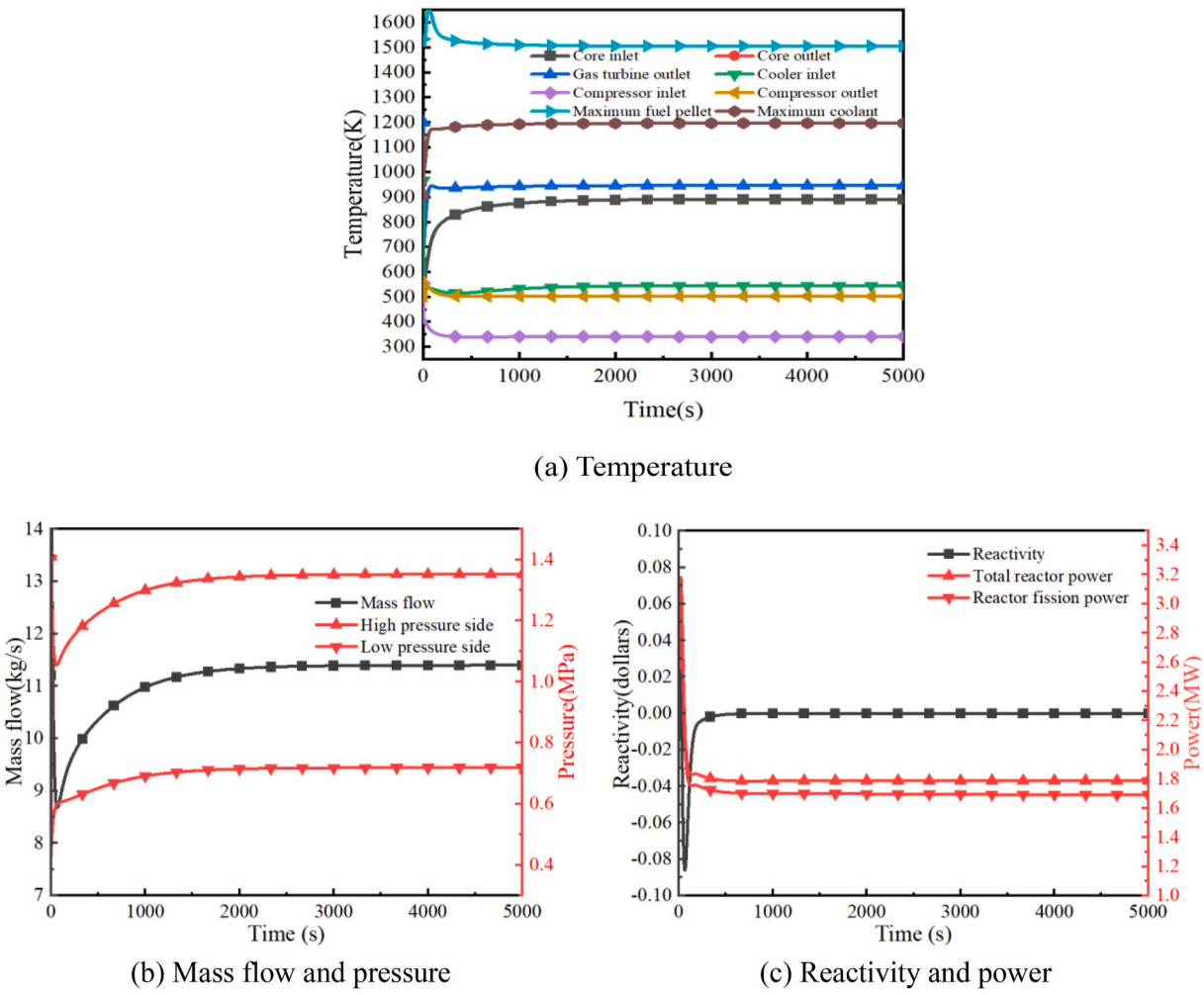


Fig. 8. Schematic diagram of key parameters changing with time.

Table 3
Verification of operating parameters of Brayton cycle system.

Variable name	Design value	Calculated value	Error
Core inlet temperature (K)	907.00	890.57	1.81%
Core inlet pressure (MPa)	1.41	1.35	4.26%
Core outlet temperature (K)	1200.00	1196.00	0.33%
Core outlet pressure (MPa)	1.40	1.35	3.78%
Gas turbine outlet temperature (K)	966.30	946.22	2.08%
Gas turbine outlet pressure (MPa)	0.73	0.72	2.18%
Outlet temperature of hot section of regenerator (K)	553.70	544.94	1.58%
Outlet pressure of hot section of regenerator (MPa)	0.70	0.71	−1.43%
Outlet temperature of hot section of cooler (K)	340.00	340.48	−0.14%
Outlet pressure of cooler hot section (MPa)	0.67	0.70	−4.35%
Compressor outlet temperature (K)	501.60	502.44	−0.17%
Compressor outlet pressure (MPa)	1.41	1.40	0.85%
Outlet temperature of cold section of regenerator (K)	907.00	896.19	1.19%
Outlet pressure of regenerator cold section (MPa)	1.41	1.37	2.49%
Mass flow (kg/s)	11.2	11.40	1.77%

2.4. Compressor model

The compressor model is similar to the pump model. Its effect on gas is the same as that of pump on single-phase and two-phase fluid. The compressor is usually driven by a turbine on the same shaft, although other functions currently available for pumps, namely tachometer, motor torque meter and sliding function, can also be used in the compressor. The compressor head curve specification is expressed as the pressure ratio, rather than directly expressed as the head as in the pump model. The compressor torque is calculated according to the compressor characteristic curve and stage insulation efficiency. Compressor components include inlet connections and control volumes, as well as optional outlet connections. The outlet joint can be a common nozzle component (such as single-volume component, time-dependent volume Component or branch component) or an inlet joint of another compressor component. This paper refers to Zhang et al. (Wright and Sanchez, 2005) method to improve the calculation model in RELAP5. The improved calculation model and implementation method are introduced below.

2.4.1. Theoretical model

In the compressor, the force acting along the tangent direction only causes the change of the angular momentum of the working fluid of the compressor. Radial force and axial force are limited by physical design. Therefore, the total torque applied to the compressor can be considered to be composed of two parts, one part changes the angular momentum of the fluid, and the other part generates reaction forces in the journal and

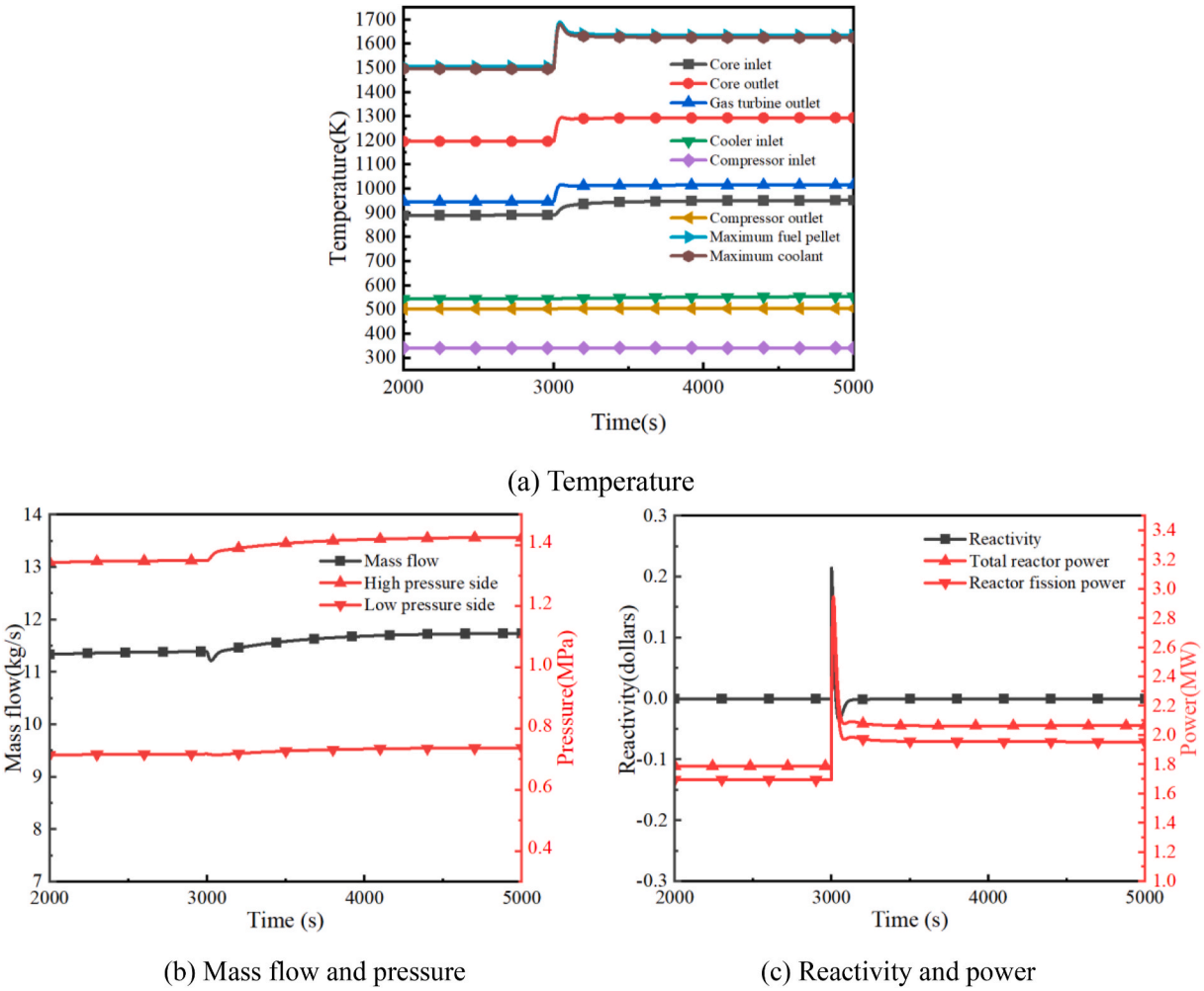


Fig. 9. Schematic diagram of key parameters changing with time.

thrust bearing. In principle, the isentropic torque can be calculated by considering the isentropic compression of the fluid, and then the total torque can be obtained by applying the efficiency factor.

The torque corresponding to isentropic work is:

$$\tau_s = \frac{\dot{m}}{\omega}\left(h_{2'}^T - h_1^T\right) \tag{14}$$

Insulation efficiency is defined as:

$$\eta_{ad} = \frac{Isentropic}{Actual} = \frac{h_{2'}^T - h_1^T}{h_2^T - h_1^T} \tag{15}$$

Irreversible or dissipated torque is:

$$\tau_d = \frac{\dot{m}}{\omega}\left(h_2^T - h_{2'}^T\right) \tag{16}$$

Compressor loss is the energy related to the work produced by friction

$$\dot{W}_d = \tau_d \cdot \omega \tag{17}$$

Compressor loss is the energy related to the work produced by friction

$$\int_{P_1}^{P_2} \frac{dP}{\rho} + \frac{v_2^2 - v_1^2}{2} = g \cdot H \tag{18}$$

Where P is the fluid pressure, ρ is the fluid density, and h is the pressure head that the compressor adds to the fluid. If the average density is assumed to be independent of pressure ($\rho_m = \frac{\rho_1 + \rho_2}{2}$), the compressor head added to the momentum equation is:

$$\Delta P = \rho \cdot g \cdot H = P_1^T(R_P - 1) = P_2^T \frac{R_P - 1}{R_P} \tag{19}$$

Assuming that the heat entering the system is zero, the total work of the input liquid is given by the following formula:

$$\begin{aligned} \dot{W}_{c.v.} &= \dot{m}\left(h_2^T - h_1^T\right) \\ \dot{W}_{c.v.} &= \dot{m}\left(h_{2'}^T - h_1^T\right) + \dot{m}\left(h_2^T - h_{2'}^T\right) \\ \dot{W}_{c.v.} &= \dot{W}_s + \dot{W}_d \end{aligned} \tag{20}$$

Isentropic work is calculated according to the increase of potential energy in the control volume

$$\dot{W}_s = \dot{m} \cdot g \cdot H = \frac{\dot{m} \cdot \Delta P}{\rho_m} \tag{21}$$

The torque corresponding to isentropic work is obtained by combining equations (14), (19) and (21):

$$\tau_s = \frac{\dot{m}}{\omega}\left(h_{2'}^T - h_1^T\right) = \frac{\dot{m}}{\omega}\frac{P_1^T(R_P - 1)}{\rho_m} \tag{22}$$

Irreversible or dissipative torque is obtained by combining equations (16) and (22):

$$\tau_d = \frac{\dot{m}}{\omega}\frac{1 - \eta_{ad}}{\eta_{ad}}\left(h_{2'}^T - h_1^T\right) = \frac{\dot{m}}{\omega}\frac{1 - \eta_{ad}}{\eta_{ad}}\frac{P_1^T(R_P - 1)}{\rho_m} \tag{23}$$

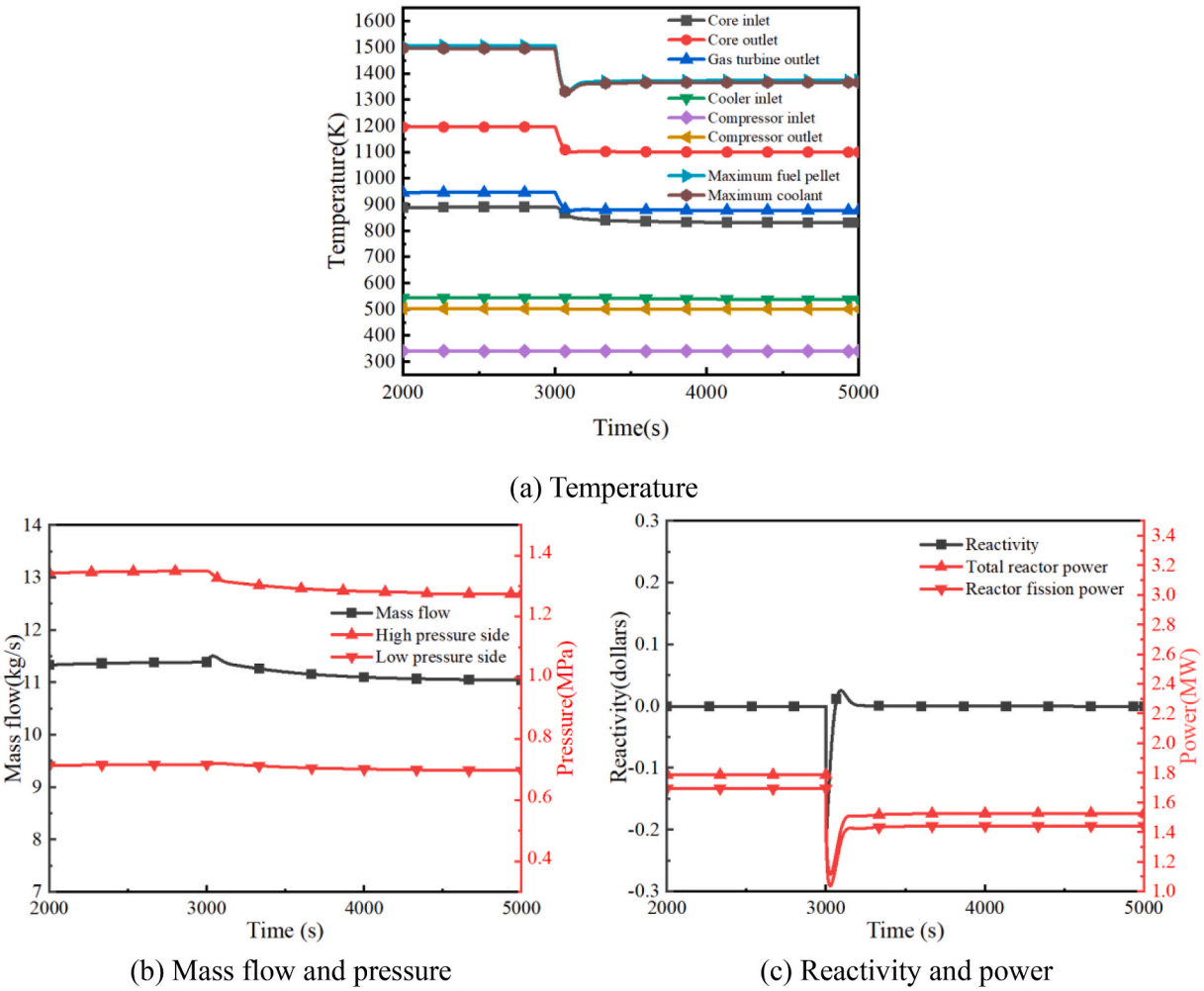


Fig. 10. Schematic diagram of key parameters changing with time.

The total torque is the sum of isentropic torque and dissipative torque

$$\tau_t = \tau_s + \tau_d = \frac{\dot{m}}{\omega}\frac{1}{\eta_{ad}}\left(h_{2'}^T - h_1^T\right) = \frac{\dot{m}}{\omega}\frac{P_1^T(R_P - 1)}{\rho_m \eta_{ad}} = \frac{\dot{m}}{\omega}\frac{(P_2 - P_1)}{\rho_m \eta_{ad}} \tag{24}$$

and it can be input into the shaft rotation speed equation.

Like the pump model, the quasi-static model of compressor performance is also applicable to volume connected flow path representation. The compressor is a volume oriented component, and the pressure head generated by the compressor is added to the connection between the compressor volume suction and the system. By assuming that the pressure head produced by the compressor is similar to body force, the compressor model is connected with the two fluid hydrodynamic model. Therefore, the first term appears in the momentum equation of the mixture; However, like gravity body force, it does not appear in the differential momentum equation used in original system analysis program. The terms added to the momentum equation of the mixture are $\rho_m gH$, where H is the total head of the pump (m), ρ_m is the bulk fluid density, and g is the gravitational acceleration.

2.4.2. *Implementation in system analysis program*

In semi implicit and near implicit numerical algorithms, the compressor head h is implicitly coupled to the velocity through its dependence on the volumetric flow rate Q. Volume flow rate is defined as the volume mass flow rate divided by the volume density. Assuming that the pressure head depends on the volume flow, it can be approximated by two Taylor series expansions, and the formula is as follows:

$$H^{n+1} = H^n + \left(\frac{dH}{dQ}\right)^n \left(Q^{n+1} - Q^n\right) \tag{25}$$

Therefore, in the two algorithms, the equivalent value of ρgh is:

$$\rho^n g H^n \Delta t + \rho^n g \left(\frac{dH}{dQ}\right)^n \left(Q^{n+1} - Q^n\right)\Delta t \tag{26}$$

Add this term to the right side of the momentum equation of the mixture.

The compressor energy dissipation Q of the compressor volume is calculated in a similar way to the turbine.

If the fluid is assumed to be incompressible, the density is constant, and the compressor dissipation is consistent with the pump dissipation. The energy dissipation term is explicitly calculated in both semi implicit and near explicit algorithms, and is distributed in the thermal energy equation of liquid and steam in a way that makes the temperature rise caused by dissipation equal at each stage. Therefore, on the right side of the liquid and steam or gas thermal energy equation of the compressor volume, the energy dissipation function (DISS) term added to the equation is:

$$DISS_f = \tau^n \omega^n \left(1 - \eta_{ad}\frac{\rho_m}{\rho_1}\right)\Delta t \cdot \left(\frac{\alpha_f^n \rho_f^n C_{pf}^n}{\alpha_f^n \rho_f^n C_{pf}^n + \alpha_g^n \rho_g^n C_{pg}^n}\right) \tag{27}$$

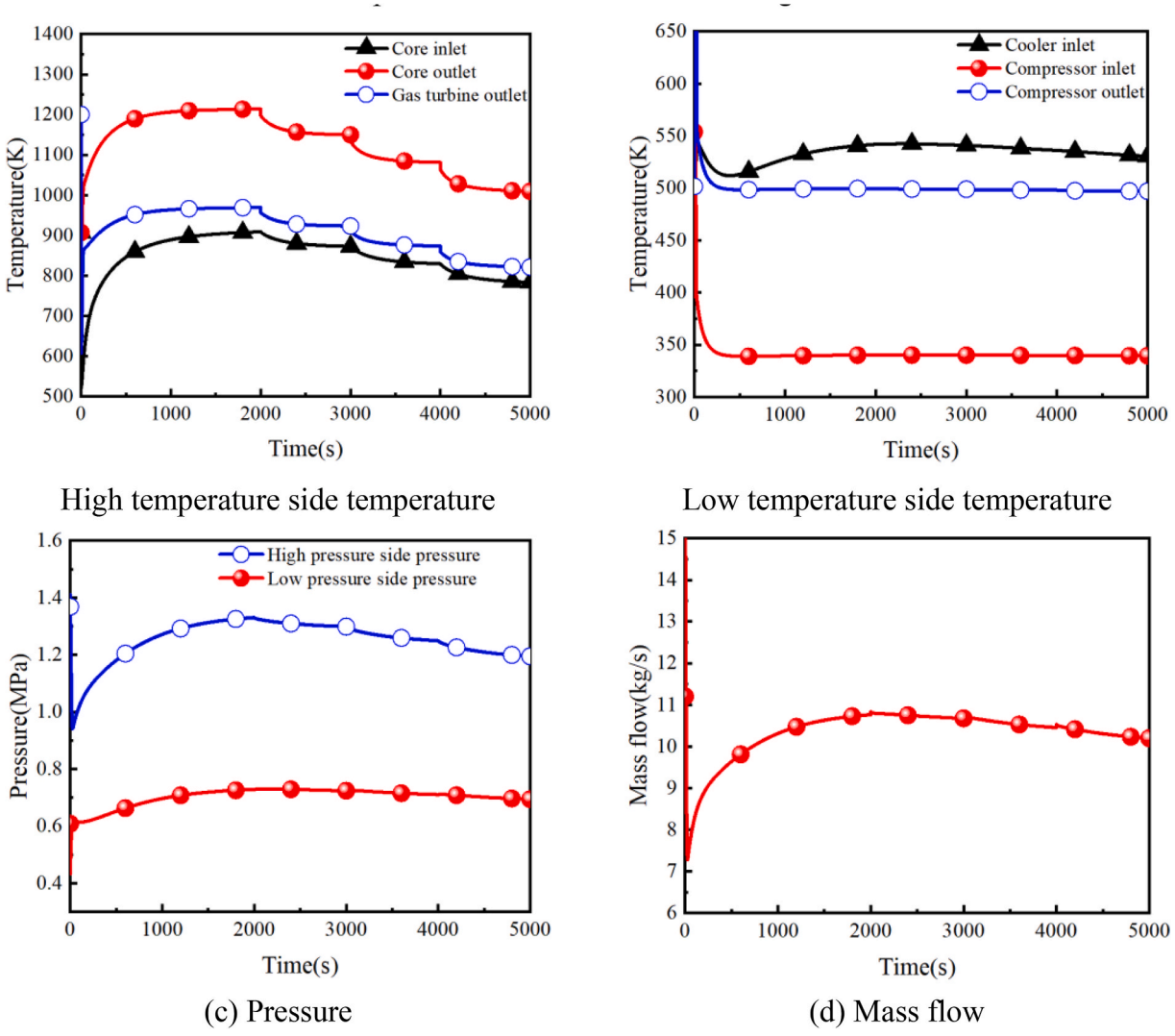


Fig. 11. Schematic diagram of key parameters changing with time.

$$DISS_g = \tau^n \omega^n \left(1 - \eta_{ad}\frac{\rho_m}{\rho_1}\right)\Delta t \cdot \left(\frac{\alpha_g^n \rho_g^n C_{pg}^n}{\alpha_f^n \rho_f^n C_{pf}^n + \alpha_g^n \rho_g^n C_{pg}^n}\right) \tag{28}$$

among α represents the volume fraction of phase, C_p is the specific heat capacity at constant pressure.

Compressor head H is defined by equation (19), torque τ defined by equation (24). The value is obtained from the compressor characteristic curve (Meng et al., 2019), and the independent variables of mass flow m and speed N, as well as the pressure ratio R_P and efficiency η the dependent variable of is entered into the code. The surge line and choke line in the compressor characteristic curve determine the normal operation range of the compressor. The data of the normal operation area of the compressor is written into input card in the form of table in the input card and read by linear interpolation method. At the same time, the transient parameters beyond the normal operating range are calculated by simple linear extrapolation. With the rated stagnation speed, rotating speed and mass flow as input parameters, the compressor characteristic curve is called to obtain the compressor pressure ratio and efficiency under the corresponding working conditions.

3. System modeling

According to the physical property calculation model and key equipment calculation model proposed above, the function of RELAP5 program is expanded and improved, and the development of simulation calculation program for He–Xe mixture gas reactor system is completed. At the same time, based on the principle diagram of the Brayton cycle system shown in Fig. 1, this paper divides the nodes of the closed Brayton cycle system of the He–Xe mixture cooled small reactor, as shown in Fig. 5.

The He–Xe mixture is heated in the reactor core. Pipe 115 is the descending section of the reactor core, and pipe 125 is the ascending section of the reactor core. The heat structure 125 wrapped around the outer wall simulates the fuel rod of the reactor core. The heated He–Xe mixed gas flows into the single volume components 121 through the single junction components 120 and then enters the turbine components 135. The working medium flowing out from the turbine enters the hot side pipe 145 of the regenerator through the single junction components 130 and 132 and the single volume components 131. The regenerator is composed of two pipe parts and a heat structure. Pipes 145 and 185 respectively represent the cold and hot sides of the regenerator, and heat exchange is conducted through the heat structure 145. The fluid flowing out from the hot side of the regenerator enters the cooler hot side pipe 155 through the single junction components 140, 040 and the single volume components 045. The cooler is composed of three pipe components and two heat structure. Pipes 255 and 355 represent the cold side of the cooler, and the internal working medium is water, which exchanges heat with the hot side through heat structure 255 and 355.245,

265, 345 and 365 are time dependent volume components, which define the inlet temperature, inlet flow rate and outlet pressure of the fluid at the cold side of the cooler, and are connected to the pipe components through single junction components 240, 250, 340 and 350. The fluid from the hot side of the cooler enters the compressor 165 through the single junction components 050, 150 and the single volume component 065. The fluid from the compressor returns to the cold side pipe 185 of the regenerator through the single junction components 160, 170 and pipe 175, and finally returns to the core descending section pipe 115 through the single junction components 180, 190 and the single volume component 195. Relevant parameters of thermal components shall be taken as per stainless steel materials.

Refer to the conceptual design parameters of JIMO and S^4 to improve the setting of design values at each key node of the helium and xenon gas cooled small reactor system. Take the design value as the initial condition, write it into the input card, calculate the steady-state operation results of the system, and compare and analyze the calculated results with the design value, the calculation results are compared with the design values, and the results are shown in Table 2. It can be seen from the table that the maximum error is 5.21%. The steady-state operation results of the built model are in good agreement with the design operation parameters, which proves the applicability of the improved simulation model.

Fig. 6 shows the schematic diagram of key parameters in Brayton cycle system changing with time when the reactor adopts point reactor dynamics. It can be seen from the figure that the inlet and outlet temperatures of other key equipment in the Brayton cycle system except the cooler can reach a stable state at about 1000s, the inlet temperature of the cooler can reach a stable state at about 2000s, and the reactor power can reach a stable state at 1000s with a small fluctuation. The inlet temperature of the cooler is affected by the outlet temperature of the gas turbine, and the outlet temperature of the cooler also affects the inlet temperature of the compressor. In order to reduce the influence of the outlet temperature of the gas turbine on the inlet temperature of the compressor, the inlet and outlet temperature of the working medium water at the cold side of the cooler is set to a constant value, which can ensure that the inlet temperature of the compressor approaches the design value as soon as possible and reduce the time for the system to reach the steady state. However, since the pressure in the system is affected by the compressor and gas turbine, it takes 2500 s to reach stability, so the mass flow in the system needs longer calculation time, and it will reach stability at about 3000 s.

According to the statistics of parameters of key nodes of Brayton cycle under different reactor powers, the steady-state response characteristics of Brayton cycle system are shown in Fig. 7. It can be found that as the power of the reactor decreases from the design power to 80% of the design power, the temperature of each node of the system gradually decreases, and the temperature of the high-temperature side of the energy conversion system decreases more than that of the low-temperature side. The change trend of pressure and temperature is consistent, and the pressure drop of high-pressure side of energy conversion system is larger than that of low-pressure side.

4. Results and discussion

4.1. Analysis of reactor feedback characteristics

It can be seen from Fig. 6 that it takes a long time for the system to achieve steady-state operation. In order to speed up the system stabilization speed, coolant temperature coefficient and fuel temperature coefficient (dollars/k) related to reactivity are added to the reactor core. Refer to (He et al., 2021a, 2021b) for specific parameters. The changes of some parameters in the Brayton cycle system after adding feedback are shown in Fig. 8. It can be seen from the figure that after adding feedback, the reactor power reaches the stable value at 200s, and the time required to reach the stable state is 800s longer than that before adding feedback. The gas turbine outlet temperature and core outlet temperature also reached stable values at 300s. Compared with the situation without feedback, the dynamics of the system reaches a stable state in a shorter time. However, due to the complexity of the calculation process of compressor and gas turbine, the pressure in the system reached a stable level in 2000s, and the d required for the calculation to reach a steady state was significantly higher than that without feedback.

When the system is stable, the parameters are shown in Table 3. In this paper, a set of design values is used for reference, so the same design values are used in Table 2 and Table 3. It can be seen from the table that after the feedback is added, the error between the calculated value and the design value of the parameters of each node in the system is smaller, with the maximum error of 4.98%, and the reactivity is 0.0 when the system operates in a steady state. Overall, after the feedback is added, the overall calculation accuracy and calculation speed of the system are improved.

4.2. Reactivity introduction accident

The accidental introduction of positive reactivity into a nuclear reactor under various design conditions leads to a sharp increase in reactor power. The causes may be: accidental withdrawal of control rods (rod lifting accident), ejection of control rods (rod ejection accident), accidental dilution of boron in coolant (uncontrolled Boron Dilution accident), and excessive cooling of main system (cold water accident). The reactivity introduction accident occurs during startup, which may lead to instantaneous criticality, and the reactor may be out of control; If the temperature in the core is too high under the power operation condition, the pressure boundary of the primary circuit system may be damaged. Bullet rod accident is a kind of accident in nuclear power plant. During the operation of PWR, the control rod drive mechanism was damaged, which led to the accident that the control rod quickly ejected from the core. This kind of accident will lead to the introduction of large reactivity into the reactor step, and the magnitude of the step introduced reactivity is equal to the integral value of the part of the ejector rod removed into the reactor.

4.2.1. Positive reactivity introduction accident

Fig. 9 is a schematic diagram of some parameters changing with time in a positive reactivity introduction accident. At 3000s, 0.21464$ reactivity was introduced into the reactor, and the increase of reactivity was equivalent to 40% of the reactivity of the control rod. It can be seen from the figure that this transient initially led to an increase in reactor power and temperature, and then with the increase in reactor inlet temperature, the reactor power decreased to respond to the increase in reactor outlet temperature. However, the reactor power will not be restored to the initial power, because the gas wheel has the opportunity to remove more energy from the gas with higher intake air temperature, resulting in an increase in system power. The turbine efficiency, mass flow and pressure drop are substantially constant.

4.2.2. Introduction of negative reactivity

Fig. 10 shows the transient response of some selected parameters when the reactor introduces negative reactivity. This transient is opposite to the introduction of positive reactivity. At 3000s, the reactivity in the reactor instantly decreased by 0.21464$, as shown in Fig. 10 (c) shows that the reactor power decreases and then slightly recovers, but it will not return to the initial power level because the gas turbine generates less power at lower temperatures. This is contrary to the effect described in the previous positive reactivity transient. The variation of power and temperature is almost the same and opposite to the positive reactive transient. All other parameters are fairly constant and conform to the objective law.

4.3. Step down load of reactor

Step load reduction is one of the transient conditions that test the load tracking ability and operation flexibility of the reactor system. In this paper, the transient simulation analysis of step load reduction for He–Xe mixture cooled small reactor system with constant circulating flow operation strategy is carried out. Control the inlet temperature of the compressor to be constant, and observe the changes of other parameters in the Brayton cycle system by reducing the reactor power. Among them, the He–Xe mixture cooled small reactor system reaches the steady state 2000s before the calculation process, and the demand is reduced by 10%–70% FP every 1000s. The whole transient process is calculated for 5000 S. The system response characteristics of small He–Xe mixture cooled reactor under step load reduction are shown in Fig. 11.

The calculation results show that the power of the gas turbine of the He–Xe mixture gas cooled reactor system decreases step by step in the whole transient process under the coordination of the controller, so as to match the required load power. At the same time, the operation of the compressor pressure controller and the cooling water flow controller ensures the quality of the He–Xe mixture working medium at the compressor and avoids the mismatch between the compressor inlet parameters and the compressor characteristic curve. Since the inlet temperature and pressure of the compressor of the system are relatively stable, the constant rotating speed of the mechanical shaft of the gas turbine makes the outlet pressure and temperature of the compressor relatively stable. However, it can be found from the calculation results of the temperature change of each key node of the system that with the step reduction of the reactor power, the circulating flow in the system also presents a step reduction trend to match the reduction of the gas turbine power, so that the temperature at the high temperature side of the energy conversion system decreases step by step.

5. Conclusion

In this paper, a semi empirical formula for predicting the physical properties of binary inert gas mixtures is used to establish a calculation model for the physical properties of He–Xe mixture, and the turbine model and compressor model in the original system analysis program are modified and improved. Thus, the function of the system analysis program is expanded and improved, and the simulation and calculation program of the He–Xe mixture cooled reactor system is developed.

1) Using the system analysis program developed in this paper, a closed Brayton cycle system with He–Xe mixture is established, and the calculated results are compared with the design values. The physical property calculation error is less than 5%, and the steady-state relative error is less than 7%. The steady-state operating results of the model are in good agreement with the designed operating parameters, which proves the applicability of the improved simulation model.
2) The steady-state response characteristics of the system were analyzed, and the parameters of each key node of the Brayton cycle under different reactor powers were statistically analyzed. The steady-state response characteristic curve of the system was obtained, and the overall change trend of the curve was consistent with the investigation results.
3) On the basis of the steady-state operation of the system, the feedback characteristic analysis of He–Xe mixture cooled small reactor was carried out. Based on the steady state operation of the system, the feedback characteristics of He–Xe hybrid cooling small reactor are analyzed. After introducing the feedback characteristic, the calculation time of reaching stability of the system is reduced by 800s compared with that without feedback, and the dynamics of the system reaches a stable state in a shorter time. After adding feedback, the error between the calculated value and the design value of each node parameter in the system is small, with the maximum error of 4.98%, which improves the calculation accuracy.
4) The developed steady-state input card is used to calculate the transient conditions such as the introduction of positive and negative reactivity and the step-down load of the reactor. The obtained trend is consistent with the existing literature research results, which verifies the feasibility of the program to calculate the transient conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work is financially supported by Postdoctoral Science Foundation of Heilongjiang Province (LBH-MZ19013) and China Postdoctoral Science Foundation (2019M650064P, 2019T120256).

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